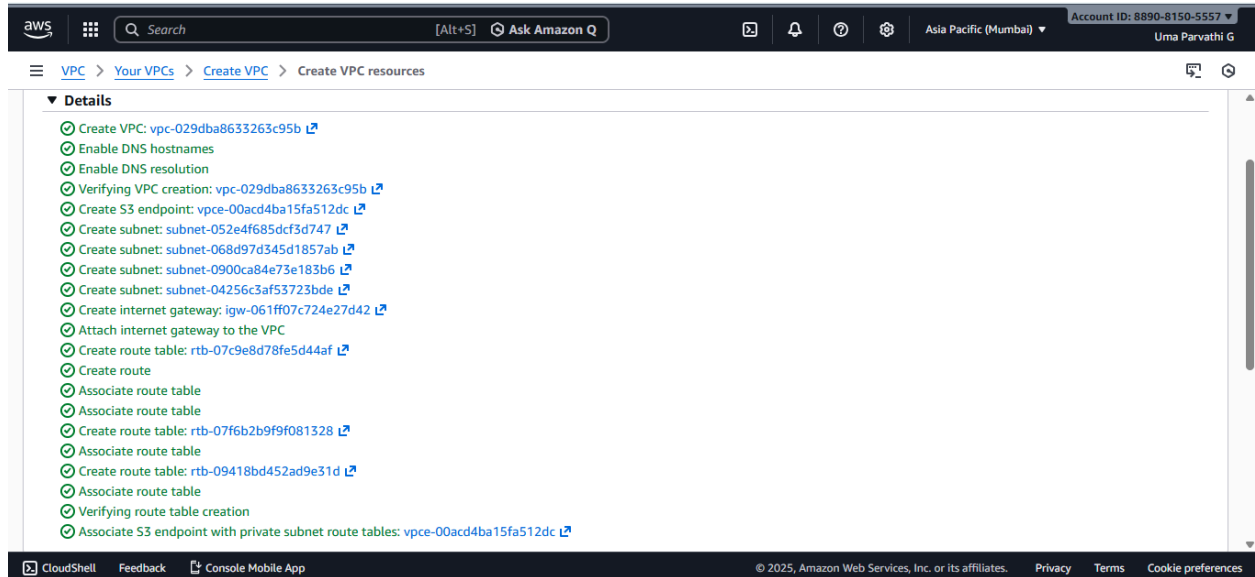
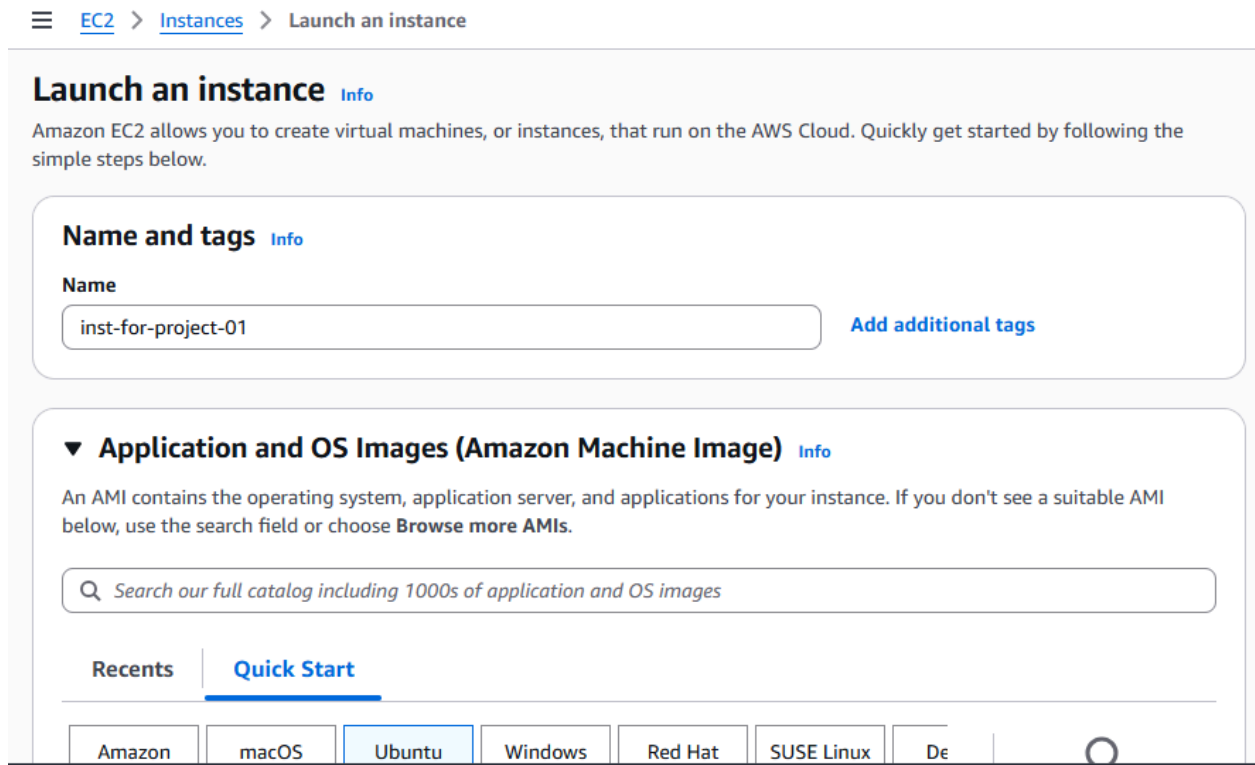


Hit create VPC



Hit view VPC

LAUNCHING INSTANCE:



In network settings:

EC2 > Instances > Launch an instance

vpc-029dba8633263c95b (project-vpc-01-vpc)
10.0.0.0/16

Subnet Info

subnet-052e4f685dcf3d747 project-vpc-01-subnet-public-1a-south-1a
VPC: vpc-029dba8633263c95b Owner: 889081505557
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.0.0/20

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Security group name - required

project-sg-01

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!#,%&()*~*'~

Description - required Info

project-sg-01 created 2025-12-24T10:28:59.999Z

Create new subnet

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more
ami-02b8269d5e85954ef

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance Preview code

EC2 > Instances > Launch an instance

Security group rule 1 (TCP, 0-65535, 0.0.0.0/0)

Remove

Type | Info

All TCP ▼

Source type | Info

Anywhere ▼

Protocol | Info

TCP

Source | Info

Port range | Info

0-65535

Description - optional | Info

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Remove

Type | Info

HTTP ▼

Source type | Info

Anywhere ▼

Protocol | Info

TCP

Source | Info

Port range | Info

80

Description - optional | Info

Add security group rule

▼ Summary

Number of instances | Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read

ami-02b8269d5e85954ef

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Hit launch instance

EC2

Dashboard
AWS Global View
Events

▼ Instances

Instances

Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ Images

AMIs

Instances (1/1) Info

Last updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☒	inst-for-projec...	i-09b9c59c9628f1bc3	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1

i-09b9c59c9628f1bc3 (inst-for-project-01)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-09b9c59c9628f1bc3	Public IPv4 address 3.110.184.231 open address	Private IPv4 addresses 10.0.1.135
IPv6 address	Instance state Running	Public DNS

CloudShell Feedback Console Mobile App

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Connecting to instance:

Hit connect

EC2 > Instances > i-09b9c59c9628f1bc3 > Connect to instance

Connect

Connect to an instance using the browser-based client.

EC2 Instance Connect Session Manager SSH client EC2 serial console

Instance ID
i-09b9c59c9628f1bc3 (inst-for-project-01)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
3.110.184.231

☐ IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

Q ubuntu X

```
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-1-135:~$
```

i-09b9c59c9628f1bc3 (inst-for-project-01)

PublicIPs: 3.110.184.231 PrivateIPs: 10.0.1.135

Installing docker:

sudo su

apt update

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-1-135:~$ ^[[200~sudo su
sudo: command not found
ubuntu@ip-10-0-1-135:~$ sudo su
root@ip-10-0-1-135:/home/ubuntu# apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
```

i-09b9c59c9628f1bc3 (inst-for-project-01)

PublicIPs: 3.110.184.231 PrivateIPs: 10.0.1.135

sudo apt install -y ca-certificates curl gnupg lsb-release

```
root@ip-10-0-1-135:/home/ubuntu# sudo apt install -y ca-certificates curl gnupg lsb-release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.3).
gnupg set to manually installed.
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 68 not upgraded.
```

Adding docker gpg:
mkdir -p /etc/apt/keyrings
curl -fsSL <https://download.docker.com/linux/ubuntu/gpg> \
| gpg --dearmor -o /etc/apt/keyrings/docker.gpg

```
root@ip-10-0-1-135:/home/ubuntu# ^C
root@ip-10-0-1-135:/home/ubuntu# ^C
root@ip-10-0-1-135:/home/ubuntu# mkdir -p /etc/apt/keyrings
root@ip-10-0-1-135:/home/ubuntu# curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
> | gpg --dearmor -o /etc/apt/keyrings/docker.gpg
curl: command not found
gpg: no valid OpenPGP data found.
root@ip-10-0-1-135:/home/ubuntu# curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
| gpg --dearmor -o /etc/apt/keyrings/docker.gpg
File '/etc/apt/keyrings/docker.gpg' exists. Overwrite? (y/N)
[1]+  Stopped                  curl -fsSL https://download.docker.com/linux/ubuntu/gpg | gpg --dearmor -o /etc/apt/keyrings/docker.gpg
root@ip-10-0-1-135:/home/ubuntu# mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
| gpg --dearmor -o /etc/apt/keyrings/docker.gpg
File '/etc/apt/keyrings/docker.gpg' exists. Overwrite? (y/N) y
```

echo \
"deb [arch=\$(dpkg --print-architecture)
signed-by=/etc/apt/keyrings/docker.gpg] \
<https://download.docker.com/linux/ubuntu> \
\$(lsb_release -cs) stable" \
| tee /etc/apt/sources.list.d/docker.list > /dev/null

```
root@ip-10-0-4-43:/home/ubuntu# echo \  
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \  
https://download.docker.com/linux/ubuntu \  
$(lsb_release -cs) stable" \  
| tee /etc/apt/sources.list.d/docker.list > /dev/null
root@ip-10-0-4-43:/home/ubuntu#
```

apt update

Docker engine installation:

apt install -y docker-ce docker-ce-cli containerd.io

```
root@ip-10-0-4-43:/home/ubuntu# apt install -y docker-ce docker-ce-cli containerd.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-buildx-plugin docker-ce-rootless-extras docker-compose-plugin libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0 pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 68 not upgraded.
Need to get 91.3 MB of archives.
After this operation, 364 MB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.2.1-1~ubuntu.24.04~noble [23.4 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.1.3-1~ubuntu.24.04~noble [16.3 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.1.3-1~ubuntu.24.04~noble [21.0 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.30.1-1~ubuntu.24.04~noble [16.4 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.1.3-1~ubuntu.24.04~noble [6383 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.0.0-1~ubuntu.24.04~noble [7709 kB]
Fetched 91.3 MB in 1s (84.7 MB/s)
Selecting previously unselected package containerd.io.
```

systemctl start docker
systemctl enable docker

Status check:
Systemctl status docker

```

root@ip-10-0-4-43:/home/ubuntu# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-12-25 16:35:36 UTC; 5min ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2604 (dockerd)
      Tasks: 9
     Memory: 42.6M (peak: 43.6M)
        CPU: 395ms
    CGroup: /system.slice/docker.service
            └─2604 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.341772211Z" level=info msg="Restoring containers: start."
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.387732286Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.394676647Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.699443450Z" level=info msg="Loading containers: done."
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.715683120Z" level=info msg="Docker daemon" commit=fbf3ed2 containerd-snapshott
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.715883578Z" level=info msg="Initializing buildkit"
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.761191962Z" level=info msg="Completed buildkit initialization"
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.769687442Z" level=info msg="Daemon has completed initialization"
Dec 25 16:35:36 ip-10-0-4-43 dockerd[2604]: time="2025-12-25T16:35:36.769742101Z" level=info msg="API listen on /run/docker.sock"
Dec 25 16:35:36 ip-10-0-4-43 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)

```

Then

usermod -aG docker ubuntu

Reconnect to instance again

Version check : docker --version

Then:

mkdir my-website

cd my-website (separately)

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# mkdir my-website
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# cd my-website
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#

```

pwd

Install git:

sudo apt install git -y

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 68 not upgraded.
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#

```

check for version:

git --version

CLONING THE GITHUB REPO INTO DIRECTORY

<https://github.com/devamithracse012-coder/frontend-aitech.git> .

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# git clone https://github.com/devamithracse012-coder/frontend-aitech.git .
Cloning into '.'...
remote: Enumerating objects: 56, done.
remote: Counting objects: 100% (56/56), done.
remote: Compressing objects: 100% (56/56), done.
remote: Total 56 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (56/56), 34.50 MiB | 20.06 MiB/s, done.
Resolving deltas: 100% (1/1), done.

```

type ls and press enter

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# ls
Bgv.mp4          bassdrum.png    drumsticks.png  exx.png         gui.jpg         login.html      minstru.jpg     review.html     tama.jpg
EXBG.png         bgvideo.mp4    effects.css      flu.jpeg        guiii.png       logo.jpg        navbar.html     roland.jpg     tuner.png
acousticdrum.png capo.png        effects.html     focusrite.png   herobg.jpeg     logos.jpeg      order.html      saxo.png       vault.png
acousticguitar.png cart.html       electronicdrum.png full.html        keey.png        midi.png        payment.html     script.js       violinn.jpg
alesis.jpg       cort.png        exa.html         gibson.png      keyyy.png       mins.jpg        piccolo.png     shop.html      wishlist.html
bass.png         dp.png         expert-guidance.html gpaylogo.png    library.html     minss.jpeg      pink.jpg        styles.css     yamaha.jpg
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#

```

Dockerizing website

1st create a docker file

vim Dockerfile



Now paste these lines

FROM nginx:alpine

COPY . /usr/share/nginx/html

EXPOSE 80


```
FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# docker build -t frontend-aitech .
[+] Building 0.2s (1/1) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 2B
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# vim Dockerfile

[5]+ Stopped vim Dockerfile
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# vim Dockerfile
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker build -t frontend-aitech .
[+] Building 10.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 97B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 76.07MB
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dddb00020ad132295
=> resolve docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dddb00020ad132295
=> sha256:da54cb821236c86af111bf7a17c40e42b3a5ad0e08c1a5ec76fb654d35f58df8 17.26MB / 17.26MB

i-0a669e194e22ae0d6 (inst-for-proj-01)
```

Press esc button, then type :wq

Then copy paste these lines

docker built -t frontend-aitech .

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker build -t frontend-aitech .
[+] Building 0.2s (1/1) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 2B
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# vim Dockerfile

[5]+ Stopped vim Dockerfile
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# vim Dockerfile
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker build -t frontend-aitech .
[+] Building 10.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 97B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 76.07MB
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dddb00020ad132295
=> resolve docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dddb00020ad132295
=> sha256:da54cb821236c86af111bf7a17c40e42b3a5ad0e08c1a5ec76fb654d35f58df8 17.26MB / 17.26MB

=> => sha256:085c5e3aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06 1.21kB / 1.21kB
=> sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9 403B / 403B
=> sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a 626B / 626B
=> sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d 953B / 953B
=> sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748f73f3c8b2d10816ae3 1.86MB / 1.86MB
=> sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB
=> extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 0.2s
=> extracting sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748f73f3c8b2d10816ae3 0.1s
=> extracting sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a 0.0s
=> extracting sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d 0.0s
=> extracting sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9 0.0s
=> extracting sha256:085c5e3aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06 0.0s
=> extracting sha256:0abf9e5672665202e79f26f23ef5dbd12358e2ea51ac32807922ab76f6b24ab0 0.0s
=> extracting sha256:de54cb821236c86af111bf7a17c40e42b3a5ad0e08c1a5ec76fb654d35f58df8 0.5s
=> [2/2] COPY . /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:8cdebef882683f16751f3d10f92c2e9ef9a51a587ccfbb0de4b13a2a086131826 0.0s
=> => exporting config sha256:b4c45b3443f0d8419f6aad0662fa40a3ede41305eb19f08db8de0c96d28ab2ad5 0.0s
=> => exporting attestation manifest sha256:2b90d5844f336646a4cec2122f97e0cf869e09f019af69bcbcdff19213bef270 0.0s
=> => exporting manifest list sha256:138a51daf56d6247fc732592021541afcf650519ef9cacc963006be2c0b6ea9e5 0.0s
=> => naming to docker.io/library/frontend-aitech:latest 0.0s
=> => unpacking to docker.io/library/frontend-aitech:latest 0.6s
```

Now type: docker images

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker images

IMAGE          ID                DISK USAGE    CONTENT SIZE  EXTRA
frontend-aitech:latest  138a51daf56d      230MB         95.4MB
```

Then run container: docker run -d -p 80:80 frontend-aitech

```

=> extracting sha256:1074353ee0db2c1d81d5af2671e36e00cf5738486f5762609ea33d606f88612 0.2
=> extracting sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748fcf73f3c8b2d10816ae3 0.1
=> extracting sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a 0.0
=> extracting sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d 0.0
=> extracting sha256:33f95a0f3229b49e777082e801b882b13f3cc5b4e389410ce8ab066f4d58c71b9 0.0
=> extracting sha256:085c5e5aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06 0.0
=> extracting sha256:0abf9e5672665202e79f26f23ef5dbd12558e2ea51ac32807922ab76fdb24ab0 0.0
=> extracting sha256:de54cb821236c86af11bf7a17c40e42b3a5ad0e08c1a5ec76fb654d35f58df8 0.5
=> [2/2] COPY . /usr/share/nginx/html 0.4
=> exporting to image 4.3
=> exporting layers 3.5
=> exporting manifest sha256:8cdeb882683f16751f3d10f92c2e9ef9a51a587ccfb0de4b13a2a086131826 0.0
=> exporting config sha256:b4ca5b3443f0d8419fead0662fa40a3ede41305eb19f08db8de0c96d28ab2ad5 0.0
=> exporting attestation manifest sha256:2b90d5844f336646a4ccc2122f97e0cf869e09f019af69bbcdfff19213bef270 0.0
=> exporting manifest list sha256:138a51daf56d6247fc732592021541afc650515ef9acc9c3006be2c0b6ea9e5 0.0
=> naming to docker.io/library/frontend-aitech:latest 0.0
=> unpacking to docker.io/library/frontend-aitech:latest 0.6
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
frontend-aitech:latest	138a51daf56d	230MB	95.4MB	

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# docker run -d -p 80:80 frontend-aitech
65fa6e0075a8b165b5673c450a9fed4e778cf020e61a82253377144f610e3863
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#

```

Check container : docker ps

```

root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
65fa6e0075a8	frontend-aitech	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:80->80/tcp, [::]:80->80/tcp	elastic_maha

```

vira
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#

```

Then type this:

mv full.html index.html

(making this full.html as home page and adding "index.html" to the name since the nginx takes only index.html as the entry page)

Then type:

docker build -t frontend-aitech-latest .

docker stop \$(docker ps -q)

docker rm \$(docker ps -aq)

docker run -d -p 80:80 frontend-aitech-latest

```

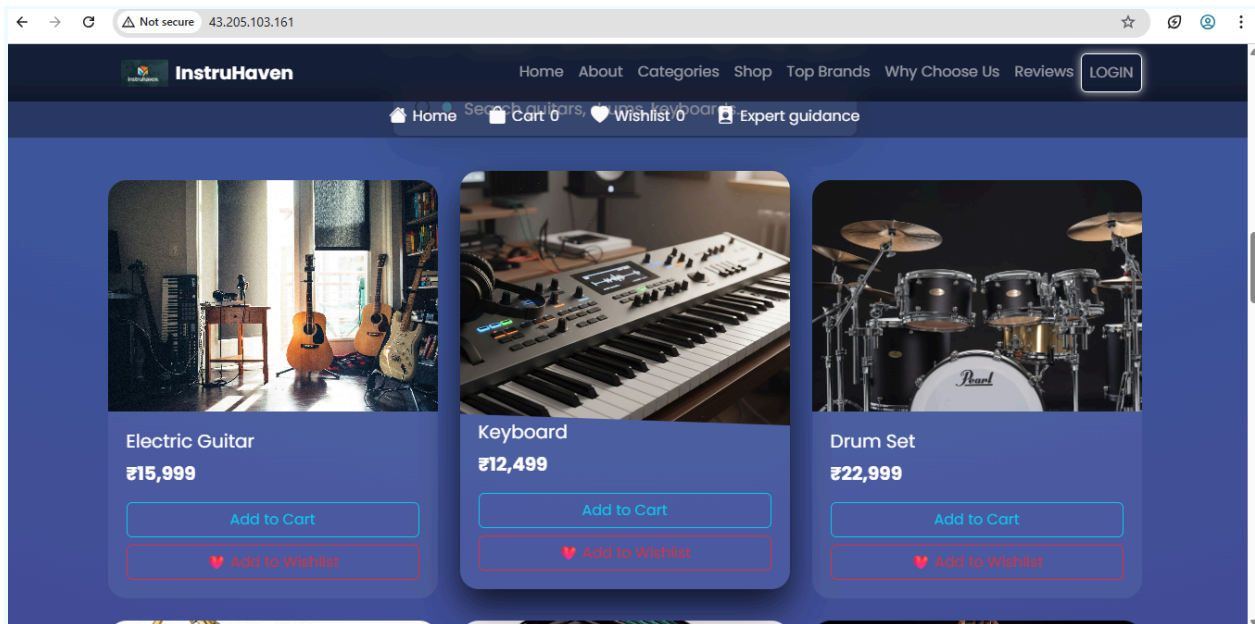
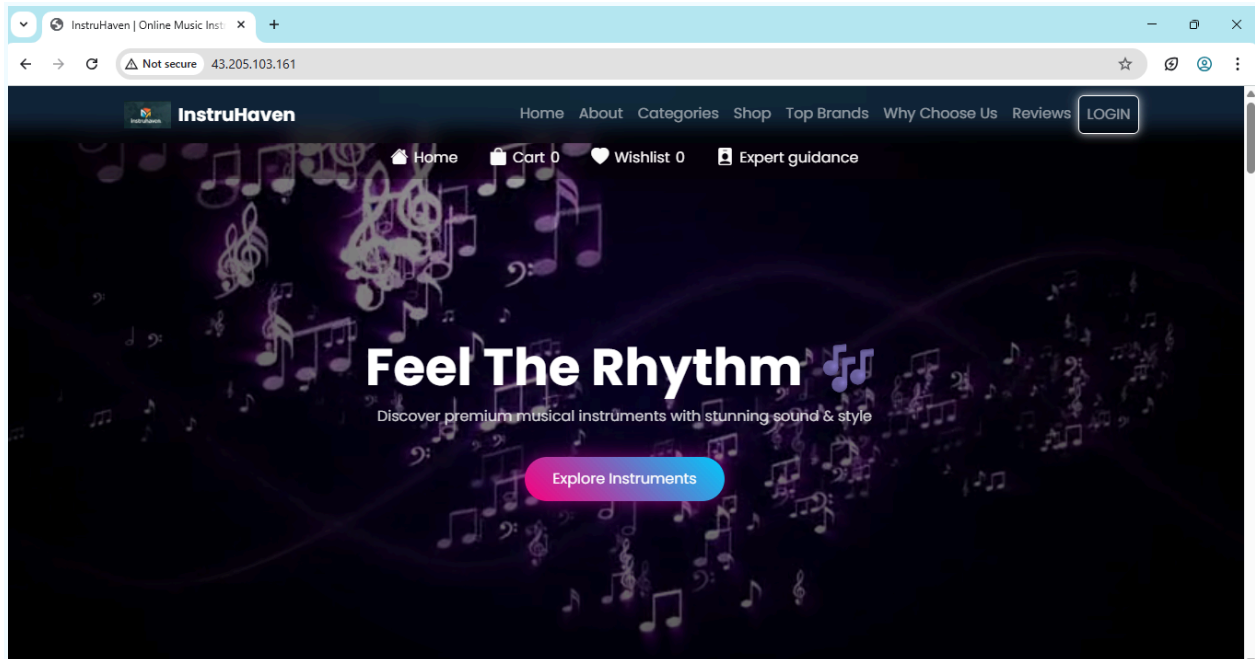
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# mv full.html index.html
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker build -t frontend-aitech-latest .
docker stop $(docker ps -q)
docker rm $(docker ps -aq)
docker run -d -p 80:80 frontend-aitech-latest
[+] Building 6.9s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 97B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> transferring context: 2B
=> [internal] load build context
=> transferring context: 53.10kB
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dd8db00020ad132295
=> resolve docker.io/library/nginx:alpine@sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dd8db00020ad132295
=> [2/2] COPY . /usr/share/nginx/html
=> exporting to image
=> exporting layers
=> exporting manifest sha256:f5917032c8737ceae39ede2ab5b486dc7aa051edf5e49231d2ef4c92573d5e3a
=> exporting config sha256:6703c229664efcad5a6dc12949d010381ebf1c87415d4680cc25e2d2856f2b37

```

Now go to new tab(better to open in new desktop)

Type : <http://43.205.103.161> (http://<public IP of EC2>)

The website will pop up



I was able to access the website

So now stop the container:
 docker stop <container-id>

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
24c99179e264   frontend-aitech-latest   "/docker-entrypoint..."  17 minutes ago   Up 17 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp   bold_noethe
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker stop 24c99179e264
24c99179e264
```

Now check whether the container had stopped

Type docker ps (this only shows currently running containers. Since we stopped it we should not get anything under the columns)

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

EXPECTED,RECEIVED.

Then type: `docker run -d -p 80:80 frontend-aitech`

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker run -d -p 80:80 frontend-aitech
0fbd1d39279ec3332b5a84a4aa6ee756547e50f9ba7d52f8504c1090efad2689
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

Again type: `docker ps`

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
0fbd1d39279e   frontend-aitech                     "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp  peaceful_noether
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

Now we push this docker image into ECR:

Containers

Amazon Elastic Container Registry

Share and deploy container software, publicly or privately

Amazon Elastic Container Registry (ECR) is a fully managed container registry that makes it easy to store, manage, share, and deploy your container images and artifacts anywhere.

Create a repository

Create

Pricing (US)

You pay only for the amount of data you store in your public or private repositories and data transferred to the Internet.

[ECR pricing](#)

Amazon ECR > Private registry > Repositories > Create private repository

Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

889081505557.dkr.ecr.ap-south-1.amazonaws.com/ frontend-aitech

15 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Image tag settings [Info](#)

Image tag mutability
Choose the tag mutability setting.

☒ **Mutable**
Image tags can be overwritten.

☐ **Immutable**
Image tags can't be overwritten.

Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

[Add filter](#)

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Amazon ECR > Private registry > Repositories

Amazon Elastic Container Service

- Private registry
 - Repositories**
 - Features & Settings
- Public registry
 - Repositories
 - Settings
- ECR public gallery [↗](#)
- Amazon ECS [↗](#)
- Amazon EKS [↗](#)
- Getting started [↗](#)
- Documentation [↗](#)

Managed signing now available
Automatically sign your container images upon push to verify authenticity and ensure supply chain security. [Configure image signing](#)

0 0 1 1 0

Private repositories (1) [View push commands](#) [Delete](#) [Actions](#) [Create repository](#)

Search by repository substring

Repository name	Repository URI copied	Created at	Tag immutability	Encryption type
☐ frontend-aitech	889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech	December 26, 2025, 00:23:46 (UTC+05.5)	Mutable	AES-256

Now install aws CLI v2 installer

curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o awscliv2.zip

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o awscliv2.zip
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 60.3M 100 60.3M 0 0 327M 0 --:--:-- --:--:-- --:--:-- 327M
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

Unzip: apt install unzip -y

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# apt install unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 68 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 0s (4852 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 71984 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes... [
Scanning processes... [
Scanning processes... [=
Scanning processes... [==
Scanning processes... [===

i-0a669e194e22ae0d6 (inst-for-proj-01)
PublicIPs: 43.205.103.161  PrivateIPs: 10.0.4.43

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Search
12:29 AM 26-12-2025
```

unzip installer: unzip awscliv2.zip

```
inflating: aws/dist/awscli/examples/dms/create-replication-task.rst
inflating: aws/dist/awscli/examples/dms/delete-endpoint.rst
inflating: aws/dist/awscli/examples/dms/stop-replication-task.rst
inflating: aws/dist/awscli/examples/dms/describe-account-attributes.rst
inflating: aws/dist/awscli/examples/dms/delete-connection.rst
inflating: aws/dist/awscli/examples/dms/describe-events.rst
inflating: aws/dist/awscli/examples/dms/describe-schemas.rst
inflating: aws/dist/awscli/examples/dms/test-connection.rst
inflating: aws/dist/awscli/examples/dms/create-endpoint.rst
inflating: aws/dist/awscli/examples/dms/modify-replication-task.rst
inflating: aws/dist/awscli/examples/dms/modify-endpoint.rst
inflating: aws/dist/awscli/examples/dms/reboot-replication-instance.rst
inflating: aws/dist/awscli/examples/dms/modify-replication-instance.rst
inflating: aws/dist/awscli/examples/codeartifact/get-package-version-readme.rst
inflating: aws/dist/awscli/examples/codeartifact/list-package-version-dependencies.rst
inflating: aws/dist/awscli/examples/codeartifact/list-repositories.rst
inflating: aws/dist/awscli/examples/codeartifact/delete-domain.rst
inflating: aws/dist/awscli/examples/codeartifact/dispose-package-versions.rst
inflating: aws/dist/awscli/examples/codeartifact/associate-external-connection.rst
inflating: aws/dist/awscli/examples/codeartifact/create-repository.rst
inflating: aws/dist/awscli/examples/codeartifact/get-authorization-token.rst
inflating: aws/dist/awscli/examples/codeartifact/get-domain-permissions-policy.rst
inflating: aws/dist/awscli/examples/codeartifact/disassociate-external-connection.rst
inflating: aws/dist/awscli/examples/codeartifact/login.rst
inflating: aws/dist/awscli/examples/codeartifact/delete-repository.rst
inflating: aws/dist/awscli/examples/codeartifact/describe-domain.rst
inflating: aws/dist/awscli/examples/codeartifact/put-repository-permissions-policy.rst

i-0a669e194e22ae0d6 (inst-for-proj-01)
PublicIPs: 43.205.103.161  PrivateIPs: 10.0.4.43

CloudShell Feedback Console Mobile App
© 2025, Amazon W
```

Install aws cli: sudo ./aws/install

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# sudo ./aws/install
You can now run: /usr/local/bin/aws --version
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

Create user:

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Specify user details

User details

User name

umag

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SignInLocal/DevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

[Cancel](#) [Next](#)

Step 1

Specify user details

Set permissions

Step 3

Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (2/1440) [Create policy](#)

Choose one or more policies to attach to your new user.

AmazonEC2ContainerRegistryFullAccess

All types

1 match

< 1 >

☒	Policy name	Type	Attached entities
☒	AmazonEC2ContainerRegistry...	AWS managed	0

Permissions policies (2/1440) [Create policy](#)

Choose one or more policies to attach to your new user.

ecs

All types

17 matches

< 1 >

☒	Policy name	Type	Attached entities
☒	AmazonECS_FullAccess	AWS managed	0
☐	AmazonECSComputeServiceR...	AWS managed	0
☐	AmazonECSInfrastructureRole...	AWS managed	0
☐	AmazonECSInfrastructureRole...	AWS managed	0
☐	AmazonECSInfrastructureRole...	AWS managed	0
☐	AmazonECSInfrastructureRole...	AWS managed	0

App

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Search

ENG IN 12:38 AM 26-12-2025

IAM > Users > Create user

Step 3
Review and create

User details

User name
umag

Console password type
None

Require password reset
No

Permissions summary

Name ↗	Type	Used as
AmazonEC2ContainerRegistryFullAccess	AWS managed	Permissions policy
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonECS_FullAccess	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

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Search

ENG IN12:39 AM26-12-2025

IAM > Users > umag

umag InfoDelete

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Summary

ARN
arn:aws:iam::889081505557:user/umag

Console access
Disabled

Access key 1
Create access key

Created
December 26, 2025, 00:39 (UTC+05:30)

Last console sign-in
-

Create access key

aws

Search

[Alt+S]

Account ID: 8890-8150-5557

Uma Parvathi G

IAM > Users > umag > Create access key

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Access key best practices & alternatives Info

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

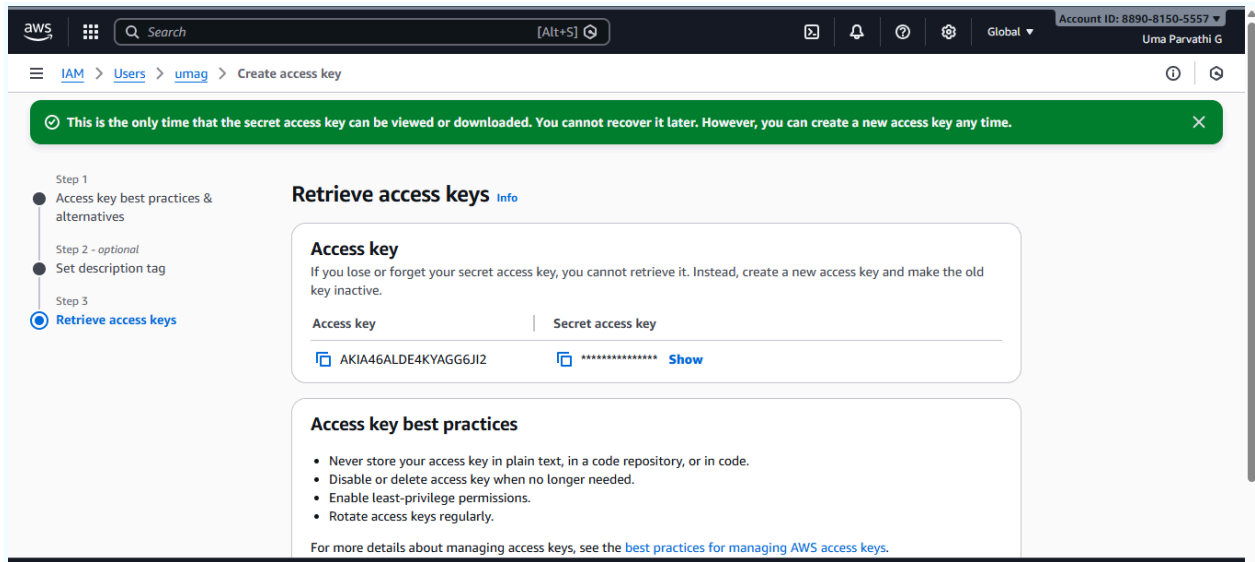
Use case

☒ Command Line Interface (CLI)
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ Local code
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ Application running on an AWS compute service
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ Third-party service
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.



Secret access key: dA4YZhBvhrJrRNA10YpVCITqvd2j0qzKR9ZTR4M2

In terminal type : aws configure

```
ubuntu@ip-10-0-4-43:~$ aws configure
AWS Access Key ID [None]: AKIA46ALDE4KYAGG6JI2
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-10-0-4-43:~$
```

Now login docker to ecr:

```
aws ecr get-login-password --region ap-south-1 \
| docker login --username AWS --password-stdin
<ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com
```

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# 889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech
bash: 889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech: No such file or directory
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# aws ecr get-login-password --region ap-south-1 \
| docker login --username AWS --password-stdin 889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

Tag Docker Image for ECR:

Type: docker images

Then tag: docker tag frontend-aitech:latest

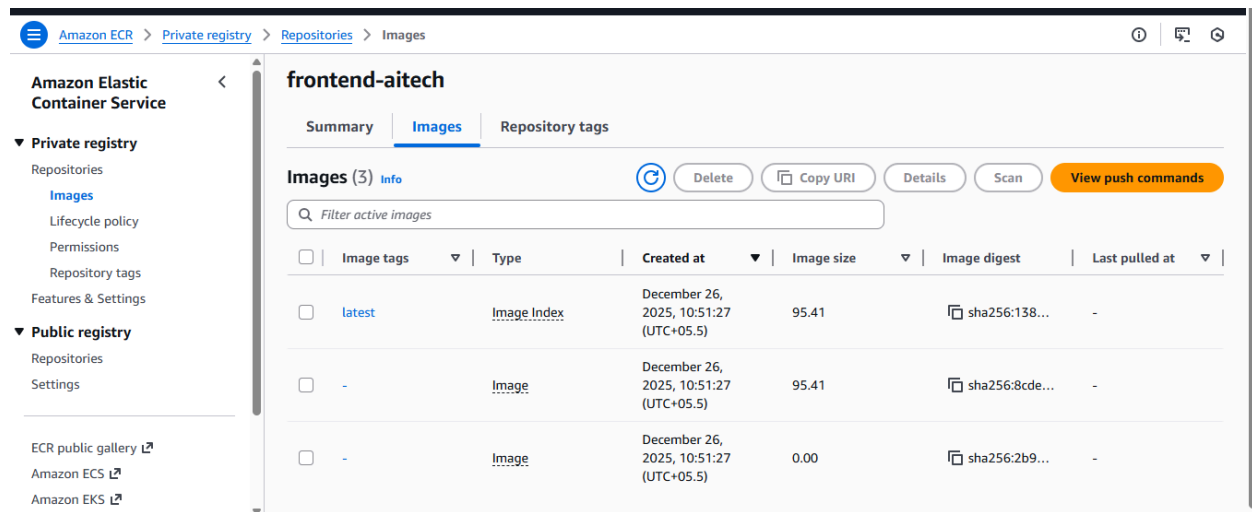
[3336xxxxxx.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech:latest](https://8336xxxxxx.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech:latest)

Push image to ecr:

Docker push <ecr_uri>:latest

```
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker tag frontend-aitech:latest 889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website# docker push 889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech:latest
The push refers to repository [889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech]
bae7218fe9a6: Pushed
c6e68efe1227: Pushed
085c5e5aaa8e: Pushed
1074353e0c0d: Pushed
0abf9a5567266: Pushed
da94cb8921236: Pushed
567f84da6fbd: Pushed
da7c973d8b92: Pushed
33f95a0f3229: Pushed
25f453064fd3: Pushed
latest: digest: sha256:138a51daf56d6247fc732592021541afc650515ef9cacc963006be2c0b6ea9e5 size: 856
root@ip-10-0-4-43:/home/ubuntu/my-website/my-website/my-website#
```

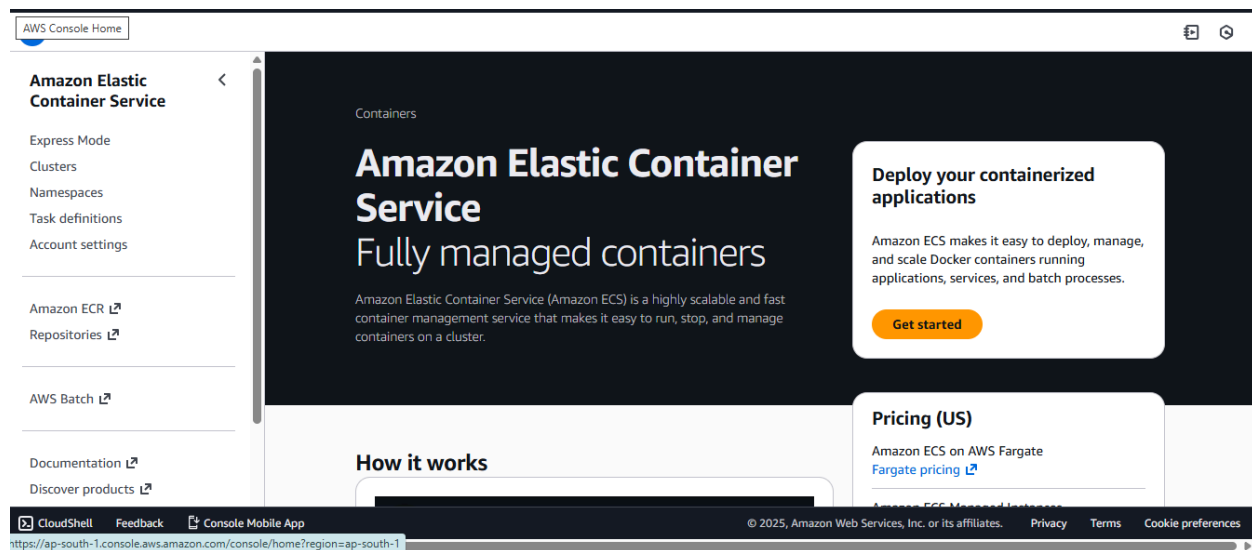
Go to ecr, at images check whether the pushed images are present are not. For me it is there.



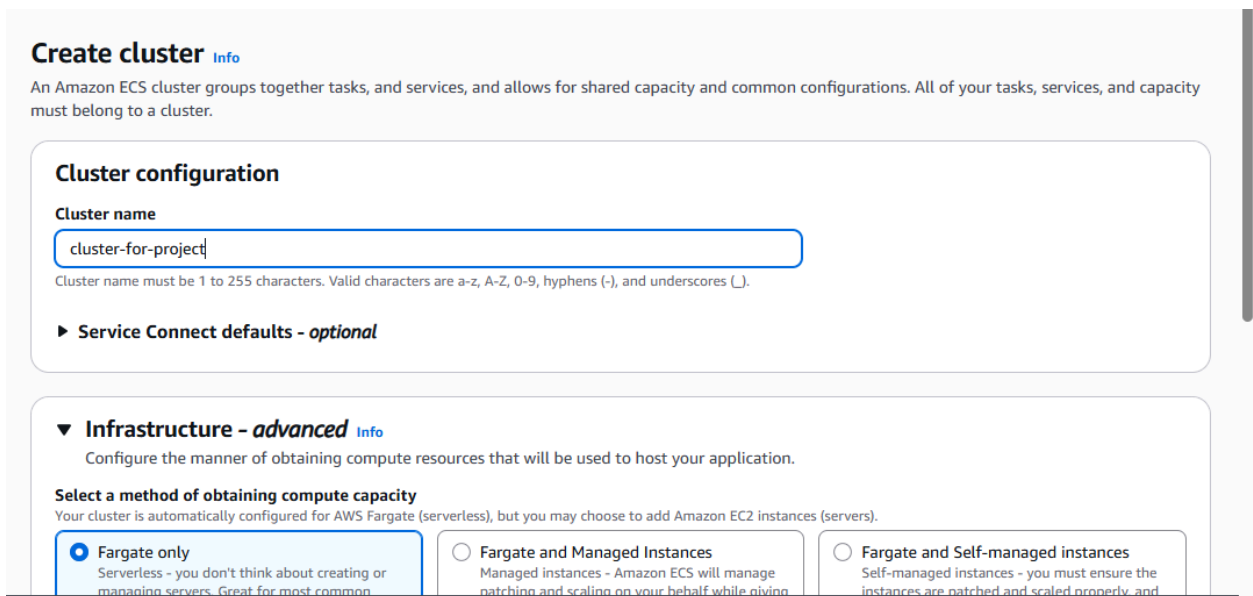
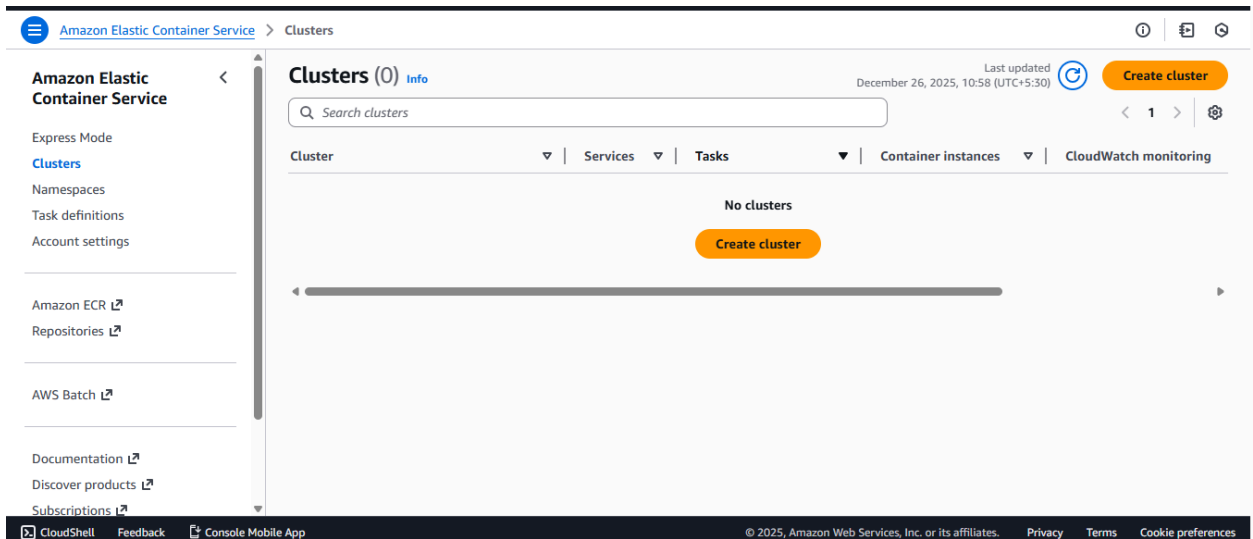
The screenshot shows the Amazon ECR console interface. On the left is a navigation sidebar with options like 'Private registry', 'Public registry', 'Repositories', and 'Settings'. The main content area is titled 'frontend-aitech' and has tabs for 'Summary', 'Images', and 'Repository tags'. The 'Images' tab is active, showing a list of three images. Each image row includes a checkbox, the image name, its type, creation date, size, digest, and last pulled date.

	Image tags	Type	Created at	Image size	Image digest	Last pulled at
☐	latest	Image Index	December 26, 2025, 10:51:27 (UTC+05.5)	95.41	sha256:138...	-
☐	-	Image	December 26, 2025, 10:51:27 (UTC+05.5)	95.41	sha256:8cde...	-
☐	-	Image	December 26, 2025, 10:51:27 (UTC+05.5)	0.00	sha256:2b9...	-

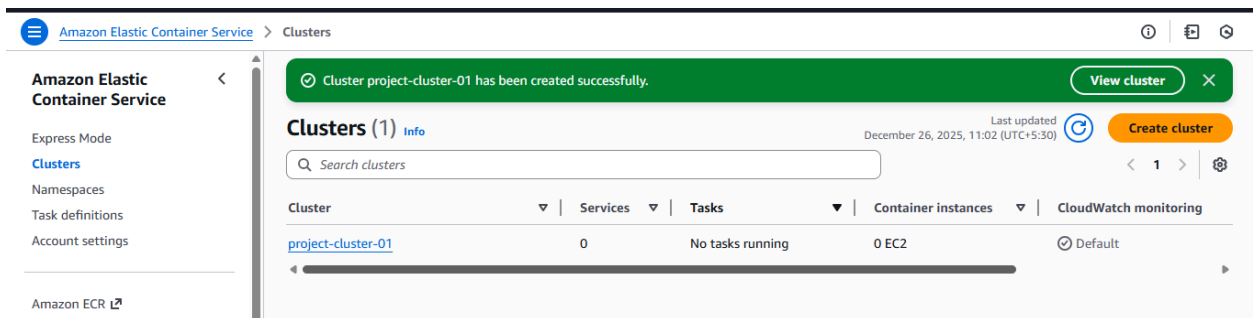
CREATE ECS CLUSTER:



The screenshot shows the AWS Console Home page for Amazon Elastic Container Service. The main heading is 'Amazon Elastic Container Service' with the subtitle 'Fully managed containers'. Below this is a brief description of ECS. To the right, there's a 'Deploy your containerized applications' section with a 'Get started' button. At the bottom, there's a 'Pricing (US)' section for Amazon ECS on AWS Fargate. The left sidebar contains navigation links for 'Express Mode', 'Clusters', 'Namespaces', 'Task definitions', 'Account settings', 'Amazon ECR', 'Repositories', 'AWS Batch', 'Documentation', and 'Discover products'.



Hit create



CREATE ECS TASK DEFINITION:

The screenshot shows the Amazon Elastic Container Service (ECS) console. The left sidebar displays the navigation menu with options like Express Mode, Clusters, Namespaces, Task definitions, Account settings, Amazon ECR, Repositories, AWS Batch, Documentation, Discover products, and Subscriptions. The main content area shows the 'project-cluster-01' cluster overview. A green banner at the top indicates that the cluster has been created successfully. The cluster details include the ARN, status (Active), CloudWatch monitoring (Default), and registered container instances. Below the overview, there are tabs for Services, Tasks, Infrastructure, Metrics, Scheduled tasks, Configuration, Event history, and Tags. The 'Services' tab is currently selected, showing a table with columns for Draining, Active, Pending, and Running states.

Cluster project-cluster-01 has been created successfully. [View cluster](#)

project-cluster-01 Last updated December 26, 2025, 11:04 (UTC+5:30) [Actions](#) [Create with Express Mode](#)

Cluster overview

ARN: [arn:aws:ecs:ap-south-1:889081505557:cluster/project-cluster-01](#) Status: Active CloudWatch monitoring: Default Registered container instances: -

Services

Draining	Active	Pending	Running
-	-	-	-

[Services](#) [Tasks](#) [Infrastructure](#) [Metrics](#) [Scheduled tasks](#) [Configuration](#) [Event history](#) [Tags](#)

Services (0) [Info](#) Last updated December 26, 2025, 11:03 (UTC+5:30) [Manage tags](#) [Update](#) [Delete service](#) [Create](#)

The screenshot shows the Amazon Elastic Container Service (ECS) console, specifically the 'Task definitions' page. The left sidebar is the same as the previous screenshot. The main content area shows the 'Task definitions (0)' page. A green banner at the top indicates that no task definitions are currently displayed. The page includes a search bar for filtering task definitions, a filter status dropdown set to 'Active', and a 'Create new task definition' button. Below the search bar, there is a table with columns for Task definition and Status of last revision. The table is currently empty, and a message states 'No task definitions to display.' with a 'Create new task definition' button.

Task definitions (0) [Info](#) Last updated December 26, 2025, 11:05 (UTC+5:30) [Deploy](#) [Create new revision](#) [Create new task definition](#)

Filter status: Active

Task definition	Status of last revision
No task definitions to display. Create new task definition	

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Create new task definition

Create new task definition

Task definition configuration

Task definition family
Specify a unique task definition family name.
td-project-01
Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Infrastructure requirements
Specify the infrastructure requirements for the task definition.
Launch type
Selection of the launch type will change task definition parameters.
☒ AWS Fargate
Serverless compute for containers.
☐ Managed Instances - new
Use if you have specific hardware constraints, such as GPU accelerators, CPU instruction sets, or network-optimized hardware (offloads scaling, patching, and instance management to AWS).

aws

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Account ID: 8890-8150-5557

Uma Parvathi G

Amazon Elastic Container Service

Create new task definition

Self-managed infrastructure using Amazon EC2 instances.

OS, Architecture, Network mode
Network mode is used for tasks and is dependent on the compute type selected.
Operating system/Architecture
Linux/X86_64
Network mode
awsipc
Task size
Specify the amount of CPU and memory to reserve for your task.
CPU
.5 vCPU
Memory
1 GB
Task roles - conditional
Task role
A task IAM role allows containers in the task to make API requests to AWS services. You can create a task IAM role from the IAM console.
-
Task execution role
A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM role created, we can create one for you.
Create default role

Amazon Elastic Container Service

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Create new task definition

Container - 1

Info

Essential container

Remove

Container details

Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

project-container-01

Essential container

Yes

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Image URI

889081505557.dkr.ecr.ap-south-1.amazonaws.com/frontend-aitech:latest

Browse ECR images

Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

Private registry

Info

Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.

☐ Private registry authentication

Port mappings

Info

Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
80	TCP	container-port-protoc	HTTP	Remove

Add port mapping

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Amazon Elastic Container Service

Create new task definition

We recommend that you use log collection for tasks running on AWS Fargate and Managed Instances. Learn more about [log collection](#).

Log collection

Info

Configure your task to send container logs to a logging destination using a default configuration. See pricing information on [Amazon CloudWatch](#).

☒ Use log collection

Destination

Amazon CloudWatch

Key	Value type	Value	
awslogs-group	Value	/ecs/td-project-01	
awslogs-region	Value	ap-south-1	
awslogs-stream-prefix	Value	ecs	
awslogs-create-group	Value	true	Remove

Add log configuration option

Hit create

The screenshot shows the AWS Elastic Container Service console. The left sidebar lists navigation options: Express Mode, Clusters, Namespaces, Task definitions (selected), Account settings, Amazon ECR, Repositories, AWS Batch, Documentation, Discover products, and Subscriptions. The main content area displays a green notification: "Task definition successfully created. td-project-01:1 has been successfully created. You can use this task definition to deploy a service or run a task." Below this, the task definition "td-project-01:1" is shown, last updated on December 26, 2025, at 11:13 (UTC+5:30). The overview section includes the following details:

ARN	Status	Time created	App environment
arn:aws:ecs:ap-south-1:889081505557:task-definition/td-project-01:1	ACTIVE	December 26, 2025, 11:13 (UTC+5:30)	Fargate

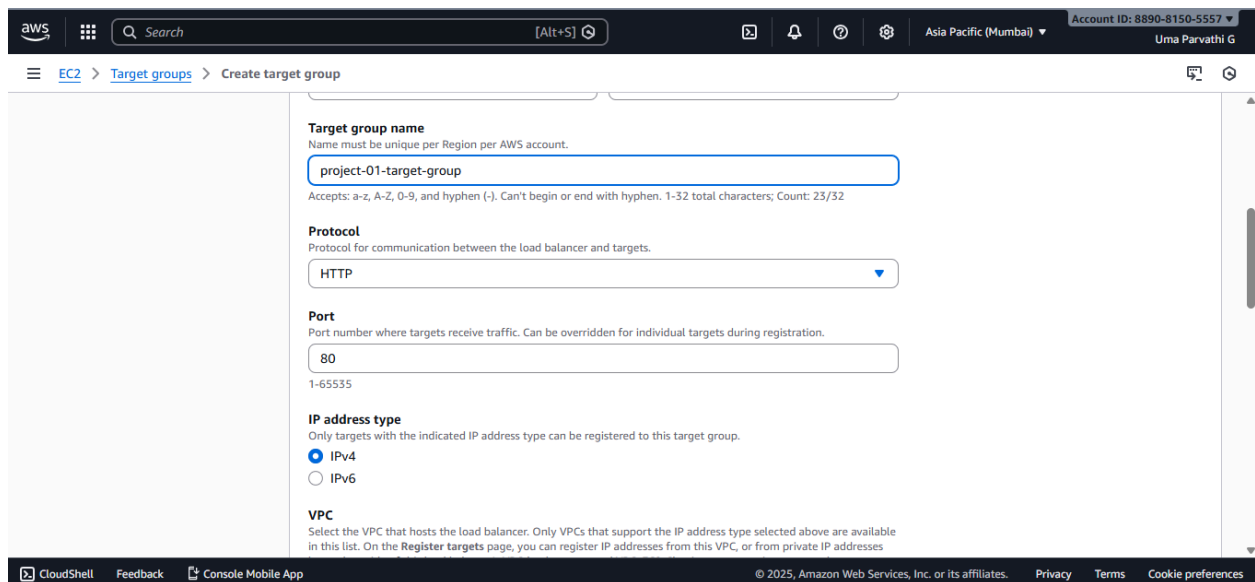
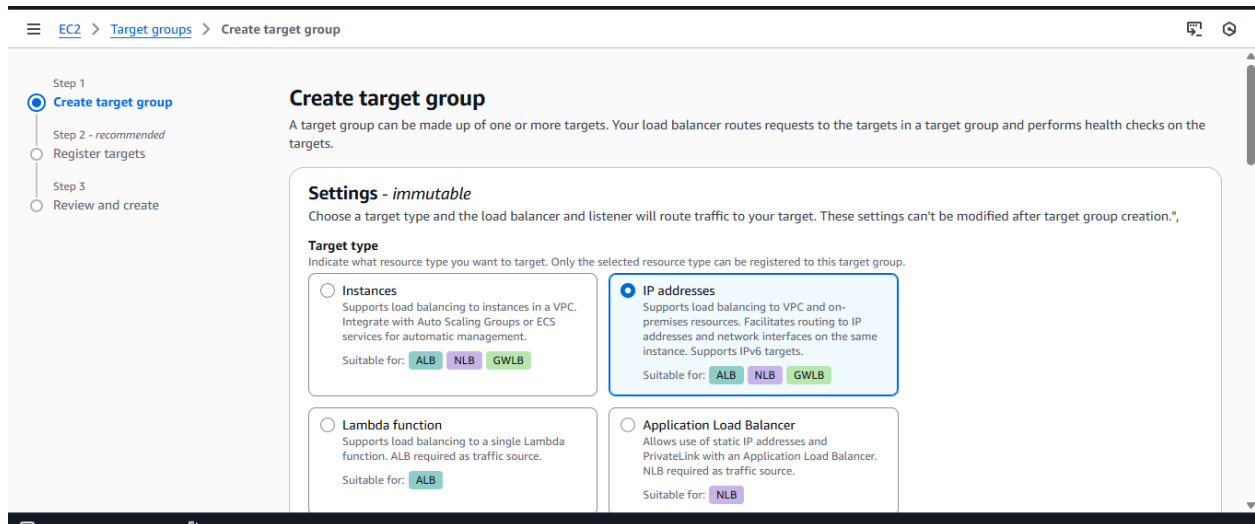
Task role	Task execution role	Operating system/Architecture	Network mode
-	ecsTaskExecutionRole	Linux/X86_64	awsvpc

Additional details: Fault injection is turned off. At the bottom, there are tabs for Containers, JSON, Task placement, Volumes (0), Requires attributes, and Tags.

CREATE TARGET GROUP:

EV2 → Target groups → load balancing → target groups

The screenshot shows the AWS EC2 console's "Target groups" page. The left sidebar lists navigation options: Snapshots, Lifecycle Manager, Network & Security (selected), Load Balancing, Auto Scaling, and Settings. The main content area shows "Target groups" with a search bar and a table with columns: Name, ARN, Port, Protocol, Target type, and Load balanc. The table is empty, with a message: "No target groups. You don't have any target groups in ap-south-1." Below the table, it says "0 target groups selected" and "Select a target group above." There is a "Create target group" button at the bottom.



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EC2 > Target groups > Create target group

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

☐ IPv6

VPC

Select the VPC that hosts the load balancer. Only VPCs that support the IP address type selected above are available in this list. On the **Register targets** page, you can register IP addresses from this VPC, or from private IP addresses located outside of this load balancer's VPC (such as a peered VPC, EC2-Classical, or on-premises targets that are reachable over Direct Connect or VPN).

vpc-025a363c28cda891f (project-vpc-01-vpc)

10.0.0.0/16

Create VPC

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

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Uma Parvathi G

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

IP addresses

Step 1: Choose a network

You can add IP addresses from the VPC selected for your target group or from outside the VPC. Note that you can assemble a mix of targets from multiple network sources by returning to this step and choosing another network.

Network

project-vpc-01-vpc

vpc-025a363c28cda891f

IPv4 VPC CIDR: 10.0.0.0/16

Step 2: Specify IPs and define ports

You can manually enter IP addresses from the selected network.

Enter an IPv4 address from a VPC subnet.

10.0.0.

Remove

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Account ID: 8890-8150-5557

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EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

IP addresses

Step 1: Choose a network

You can add IP addresses from the VPC selected for your target group or from outside the VPC. Note that you can assemble a mix of targets from multiple network sources by returning to this step and choosing another network.

Network

project-vpc-01-vpc

vpc-025a363c28cda891f

IPv4 VPC CIDR: 10.0.0.0/16

Step 2: Specify IPs and define ports

You can manually enter IP addresses from the selected network.

Enter an IPv4 address from a VPC subnet.

10.0.0.

Remove

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EC2 > Target groups > Create target group

Step 1: Create target group
Step 2 - recommended: Register targets
Step 3: Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

[Edit](#)

Target group details			
Name project-01-tg	Target type IP	Protocol : Port HTTP: 80	Protocol version HTTP1
VPC vpc-025a363c28cda891f	IP address type IPv4		

Health check details			
Health check protocol HTTP	Health check path /	Health check port traffic-port	Interval 30 seconds
Timeout	Healthy threshold	Unhealthy threshold	Success codes

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EC2 > Target groups > project-01-tg

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

Settings

Successfully created the target group: **project-01-tg**. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the **Targets** tab. [Give feedback](#)

project-01-tg

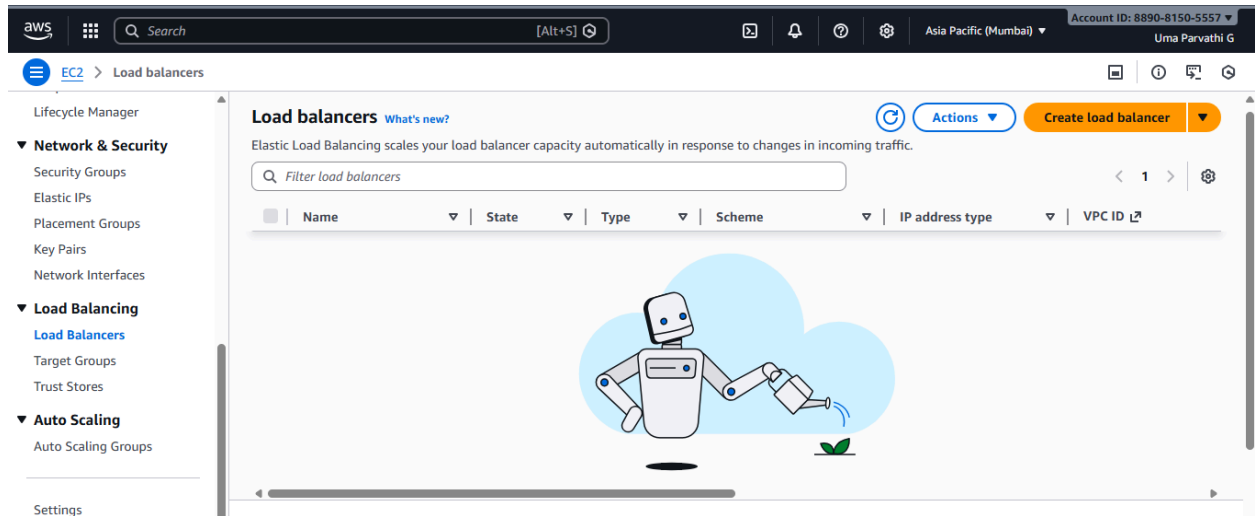
[Actions](#)

Details					
arn:aws:elasticloadbalancing:ap-south-1:889081505557:targetgroup/project-01-tg/d7323c08bea32266					
Target type IP	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-025a363c28cda891f		
IP address type IPv4	Load balancer None associated				
0 Total targets	0 Healthy	0 Unhealthy	0 Unused	0 Initial	0 Draining
0 Anomalous					

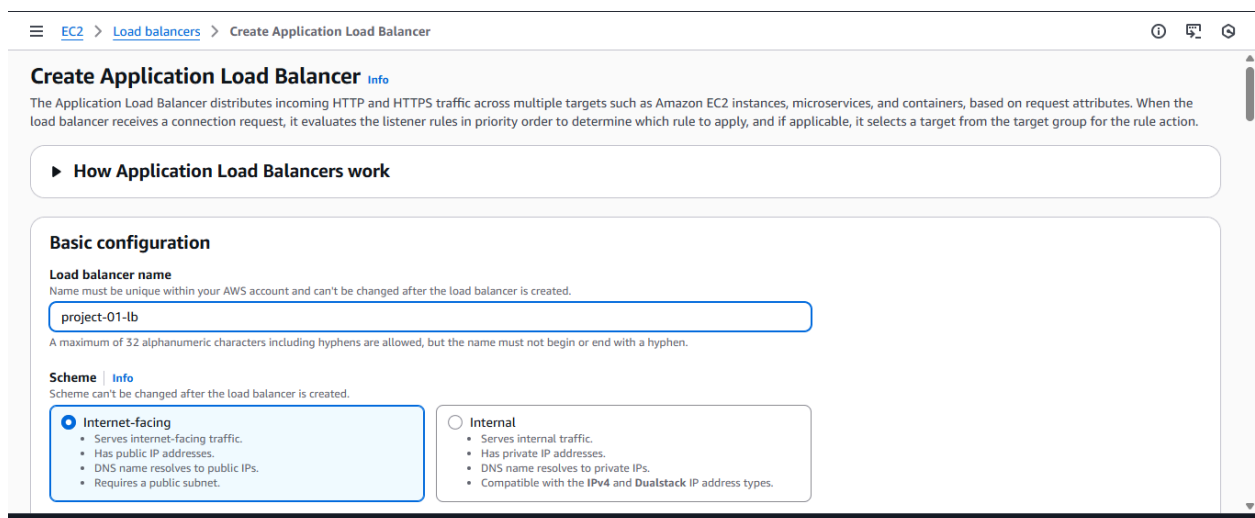
[Targets](#) [Monitoring](#) [Health checks](#) [Attributes](#) [Tags](#)

CREATE Application Load Balancer:

EC2 → Load balancer



Create load balancers → application load balancer



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Account ID: 8890-8150-5557

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EC2

Load balancers

Create Application Load Balancer

Network mapping

info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-025a363c28cda891f (project-vpc-01-vpc)

10.0.0.0/16

Create VPC

IP pools

Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-059a4a446f8b735d3

IPv4 subnet CIDR: 10.0.0.0/20

project-vpc-01-subnet-public1-ap-south-1a

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EC2

Load balancers

Create Application Load Balancer

Availability Zones and subnets

Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-059a4a446f8b735d3

IPv4 subnet CIDR: 10.0.0.0/20

project-vpc-01-subnet-public1-ap-south-1a

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-01c6bde408df30405

IPv4 subnet CIDR: 10.0.144.0/20

project-vpc-01-subnet-private2-ap-south-1b

The selected subnet does not have a route to an internet gateway. This means that your load balancer will not receive internet traffic.

You can proceed with this selection; however, for internet traffic to reach your load balancer, you must update the subnet's route table in the [VPC console](#).

Security groups

Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

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EC2 > Load balancers > Create Application Load Balancer

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

default sg-09969a013077a86af VPC: vpc-025a363c28cda891f

project-security-01 sg-09b99f8620f344393 VPC: vpc-025a363c28cda891f

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol HTTP

Port 80

1-65535

Default action Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

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EC2 > Load balancers > Create Application Load Balancer

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol HTTP

Port 80

1-65535

Default action Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group Info

Choose a target group and specify routing weight or [create target group](#).

Target group project-01-tg

HTTP

Weight 1

Percent 100%

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration Edit

Name: project-01-lb

Scheme: Internet-facing

IP address type: IPv4

Network mapping Edit

VPC: [vpc-025a363c28cda891f](#)

Public IPv4 IPAM pool: -

Availability Zones and subnets:

- ap-south-1a
 - [subnet-059a4a446f8b735d3](#)
 - project-vpc-01-subnet-public1-ap-south-1a
- ap-south-1b
 - [subnet-01c6bde408df30405](#)
 - project-vpc-01-subnet-private2-ap-south-1b

Security groups Edit

default [sg-09969a013077a86af](#)

project-security-01 [sg-09b99f8620f344393](#)

Listeners and routing Edit

HTTP:80 | [Forward to 1 target group](#)

Service Integrations Edit

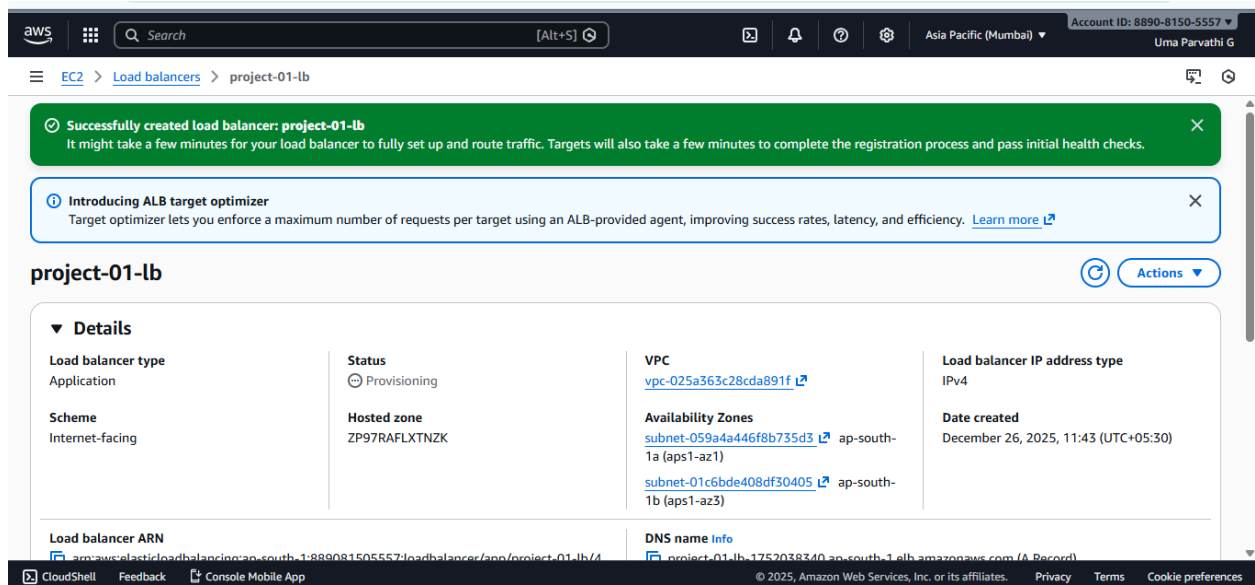
Amazon CloudFront + AWS Web Application Firewall (WAF): -

AWS WAF: -

Tags Edit

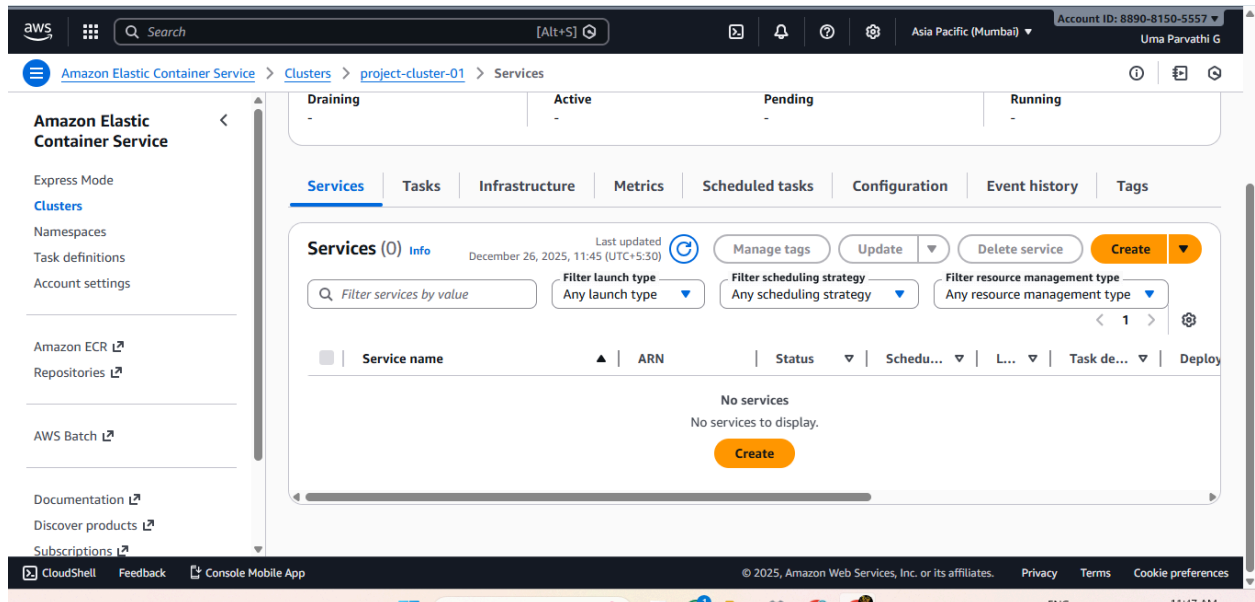
-

Hit create



CREATE ECS SERVICE AND ATTACH ALB:

ECS → clusters → click the cluster created → create services



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Amazon Elastic Container Service

Clusters

project-cluster-01

Create service

Amazon Elastic Container Service

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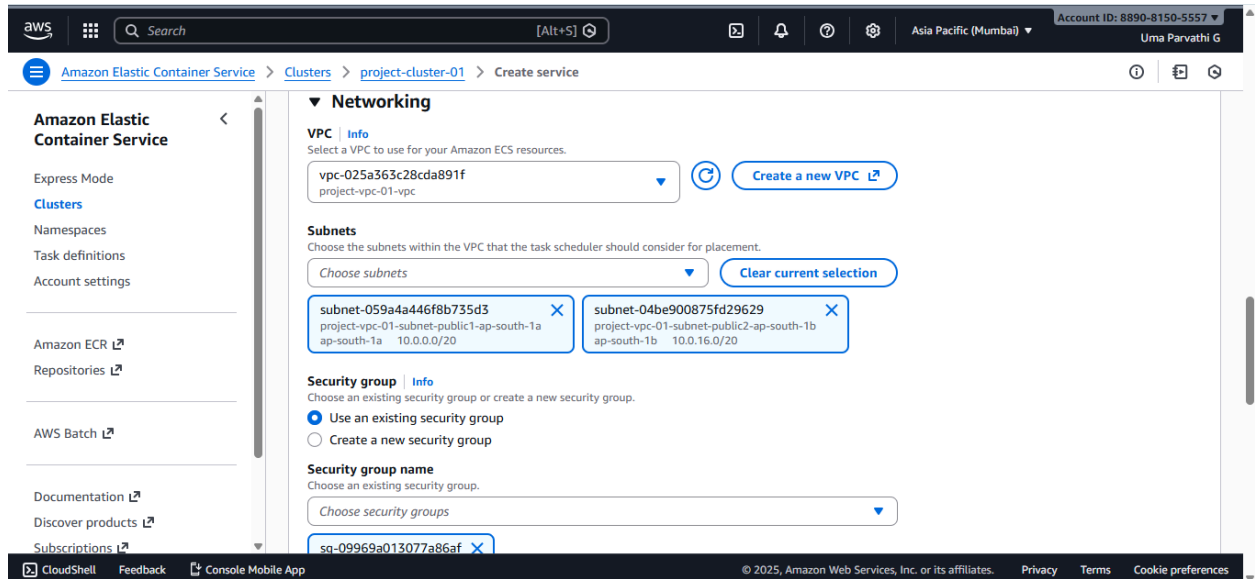
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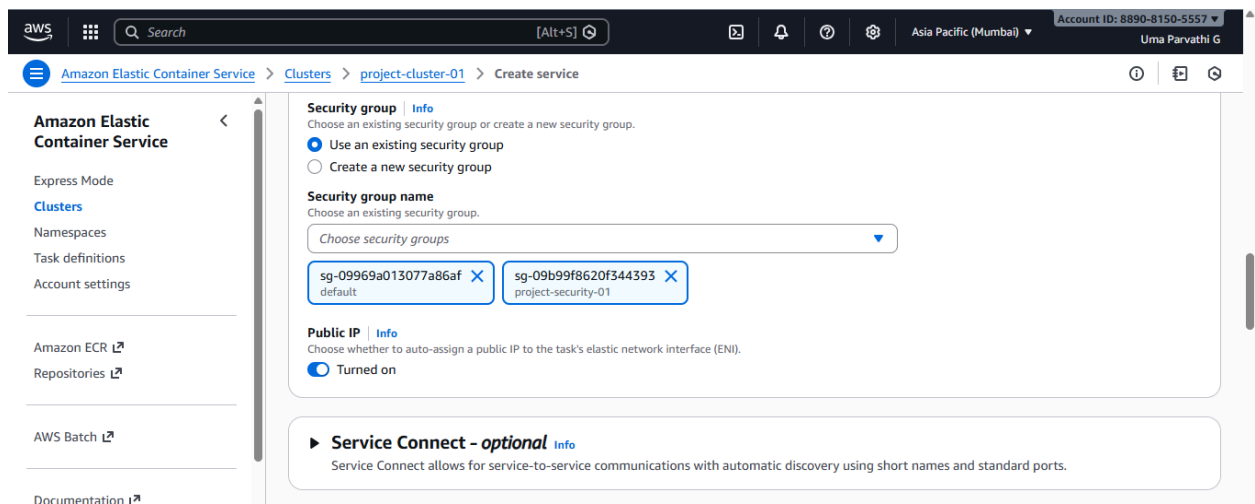
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(remove private subnets)



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Amazon Elastic Container Service

Clusters

project-cluster-01

Create service

Amazon Elastic Container Service

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VPC

The VPC for your load balancing resources must be the same as the VPC for your service with awsvpc.

vpc-025a363c28cda891f

Load balancer type

Specify the load balancer type to distribute incoming traffic across the tasks running in your service.

Application Load Balancer

An Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.

Network Load Balancer

A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).

Container

The container and port to load balance the incoming traffic to

project-container-01 80:80

Host port:Container port

Application Load Balancer

Specify whether to create a new load balancer or choose an existing one.

Create a new load balancer

Use an existing load balancer

Load balancer

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Amazon Elastic Container Service

Clusters

project-cluster-01

Create service

Amazon Elastic Container Service

Express Mode

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Load balancer

Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in EC2 Console.

project-01-lb

project-01-lb-1752038340.ap-south-1.elb.amazonaws.com

internet-facing

Listener

Specify the port and protocol that the load balancer will listen for connection requests on.

Create new listener

Use an existing listener

Listener

HTTP:80

Listener rules for 80:HTTP (1)

Traffic received by the listener is routed according to its rules. Rules are evaluated in priority order, from the lowest value to the highest value. The default rule is evaluated last.

Priority

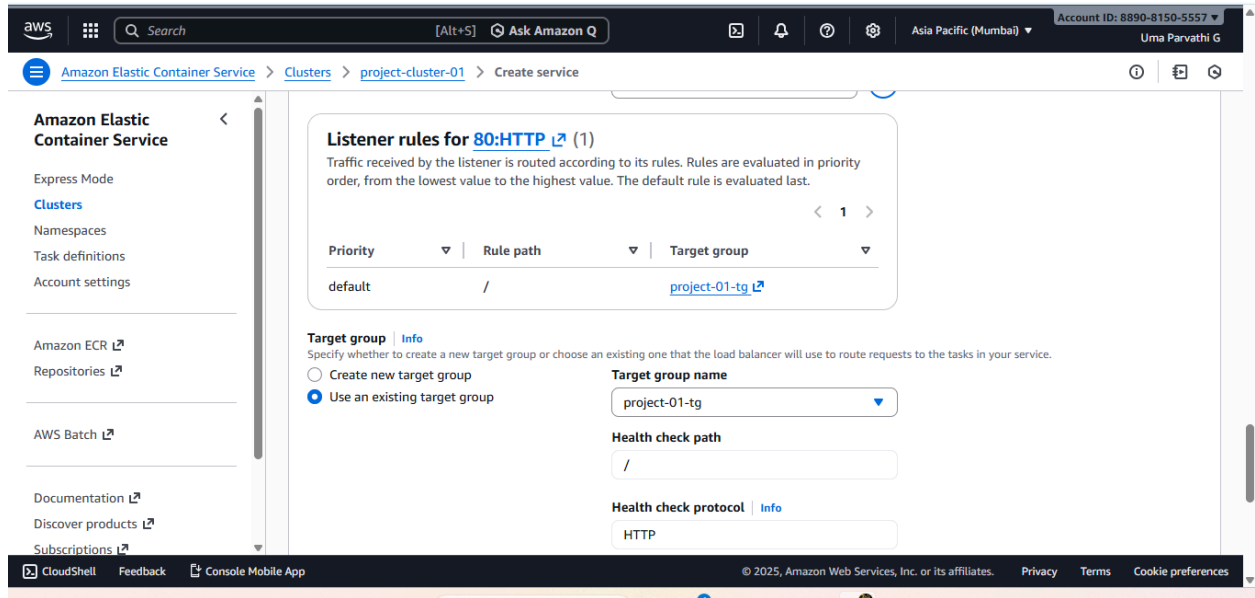
Rule path

Target group

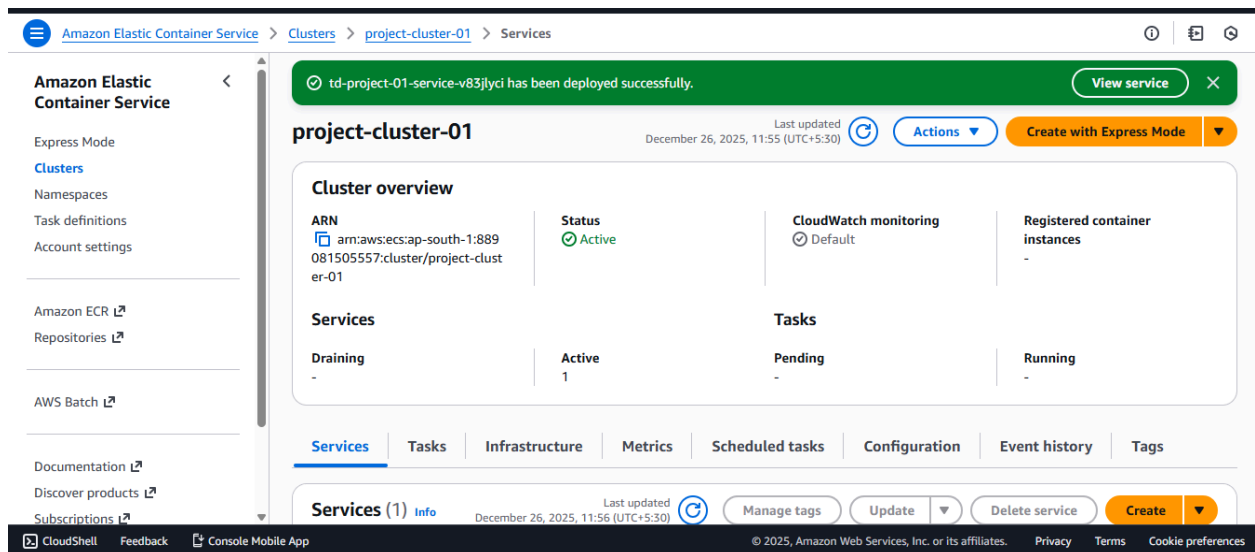
default / project-01-tg

Target group

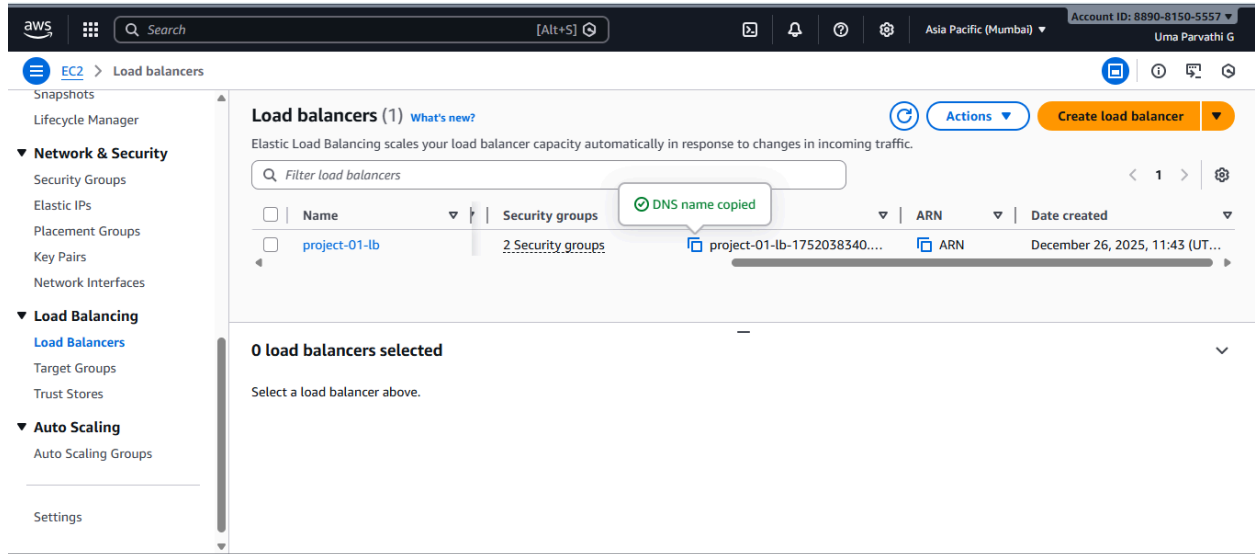
Specify whether to create a new target group or choose an existing one that the load balancer will use to route requests to the tasks in your service.



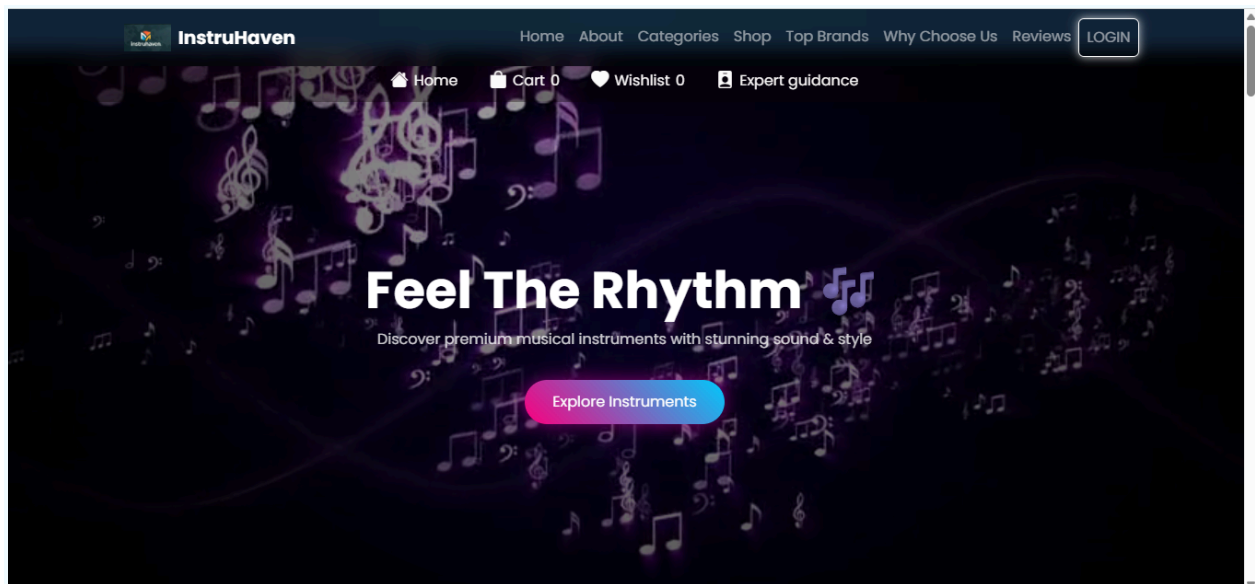
Hit create



Now go to EC2 → load balancer
Copy DNS name



Paste the DNS name in a new window/incognito tab.





Electric Guitar

₹15,999

Add to Cart

♥ Add to Wishlist



Keyboard

₹12,499

Add to Cart

♥ Add to Wishlist



Drum Set

₹22,999

Add to Cart

♥ Add to Wishlist